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AIM: Implementation of nesC Programming Model

Create a basic nesC application using modules and configurations to understand component wiring and interface binding.

Step 1: The Module (BlinkC.nc)

This component contains only the code logic. It does not know where the timer or LEDs come from; it just declares that it needs them.

CODE:

```
#include "Timer.h"

module BlinkC {
  // Interfaces used by this module
  uses interface Boot;
  uses interface Leds;
  uses interface Timer<TMilli> as Timer0;
}

implementation {
  event void Boot.booted() {
    // Command: start timer
    call Timer0.startPeriodic(500);
  }

  event void Timer0.fired() {
    // Command: toggle LED
    call Leds.led0Toggle();
  }
}
```

Step 2: The Configuration / Wiring (BlinkAppC.nc)

This component has no logic. Its only job is component wiring (binding

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interfaces together).

Code:

```
configuration BlinkAppC {  
}  
implementation {  
    // 1. Declare the components  
    components MainC;  
    components LedsC;  
    components new TimerMilliC() as Timer0;  
    components BlinkC as App;  
  
    // 2. Wire the interfaces (Interface Binding)  
    App.Boot -> MainC.Boot;  
    App.Leds -> LedsC;  
    App.Timer0 -> Timer0;  
}
```

Step 3: The Runner Script (sim.py)

CODE:

```
import sys  
from TOSSIM import *  
  
t = Tossim([])  
t.addChannel("LedsC", sys.stdout)  
  
# Boot mote 1  
m = t.getNode(1)  
m.bootAtTime(0)  
  
# Run 20 events  
for i in range(20):  
    t.runNextEvent()
```

OUTPUT:

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```
Activities terminal Wed 03:44
aquib@ubuntu: ~/tinys-main/apps/Blink

[INFO] writing XML schema to app.xml
[INFO] compiling BlinkAppC to object file sim.o
nesc -c -shared -fPIC -o simbuild/micaz/sim.o -g -O0 -gcc=gcc -Wnesc-all -fnesc-include=tos -fnesc-scheduler=TinySchedulerC,TinySchedulerC.TaskBasic,TaskBasic,TaskBasic,runTask,pos
tTask -fnesc-cfile=simbuild/micaz/app.c -fnesc-separator=_ -DTOSSIM -fnesc-nido-tosnodes=1000 -fnesc-simulate -fnesc-nido-notenumber=sim_node\(\) -fnesc-gcc=gcc -I../tos/lib/tos
sim -D.Float128=double -D.Float64=double -D.Float32=double -D.Float64x=double -D.Float32x=double -I../tos/platforms/micaz/sim -I../tos/platforms/micaz -I../tos/platforms/mi
ca/sim -I../tos/platforms/mica -I../tos/platforms/micaz/chips/cc2420/sim -I../tos/platforms/micaz/chips/cc2420 -I../tos/chips/cc2420/sim -I../tos/chips/cc2420 -I../to
s/chips/cc2420/alarm -I../tos/chips/cc2420/control/sim -I../tos/chips/cc2420/control -I../tos/chips/cc2420/csma/sim -I../tos/chips/cc2420/csma -I../tos/chips/cc242
0/interfaces/sim -I../tos/chips/cc2420/interfaces -I../tos/chips/cc2420/link/sim -I../tos/chips/cc2420/link -I../tos/chips/cc2420/lowpan/sim -I../tos/chips/cc2420/low
pan -I../tos/chips/cc2420/lpl -I../tos/chips/cc2420/packet/sim -I../tos/chips/cc2420/packet -I../tos/chips/cc2420/receive/sim -I../tos/chips/cc2420/receive -I../to
s/chips/cc2420/spl -I../tos/chips/cc2420/transmit -I../tos/chips/cc2420/unique/sim -I../tos/chips/cc2420/unique -I../tos/platforms/mica2/chips/at45db -I../tos/platfor
ms/mica/chips/at45db -I../tos/chips/at45db -I../tos/chips/atm128/sim -I../tos/chips/atm128 -I../tos/chips/atm128/adc -I../tos/chips/atm128/pins/sim -I../tos/chips/
atm128/pins -I../tos/chips/atm128/spl/sim -I../tos/chips/atm128/spl -I../tos/chips/atm128/timer/sim -I../tos/chips/atm128/timer -I../tos/lib/power -I../tos/lib/tim
er -I../tos/lib/serial -fnesc-no-debug -DPLATFORM_MICAZ -Wall -Wshadow -finline-limit=100000 -I../tos/system -I../tos/types -I../tos/interfaces -DIDENT_APPNAME="BlinkAp
pC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DIDENT_TIMESTAMP=0x6ac61fb3L -DIDENT_UIDHASH=0x05c16db5L -Wno-nesc-data-race BlinkAppC.nc
-fnesc-dump=components -fnesc-dump=variables -fnesc-dump=constants -fnesc-dump=typedefs -fnesc-dump=interfacedefs -fnesc-dump=tags -fnesc-dumpfile=app.xml
../tos/lib/tossin/sim_noise.c: In function 'makeNoiseModel':
../tos/lib/tossin/sim_noise.c:440:78: warning: cast from pointer to integer of different size [-Wpointer-to-int-cast]
    dbg("Insert", "Setting history %i to be %i\n", (int)i, (int)noiseData[node_id].key[i]);
                                                                    ^
../tos/lib/tossin/sim_noise.c: In function 'sim_noise_gen':
../tos/lib/tossin/sim_noise.c:307:7: warning: variable 'noiseIndex' set but not used [-Wunused-but-set-variable]
    int noiseIndex = 0;
    ^
../tos/lib/tossin/sim_lqi.c: At top level:
../tos/chips/atm128/sim/atm128hardware.h:101:3: warning: '__nesc_disable_interrupt' is static but used in inline function '__nesc_atomic_start' which is not static
    __nesc_disable_interrupt();
    ^
[INFO] compiling Python support and C libraries into pytosin.o, tossin.o, and c-support.o
g++ -c -shared -fPIC -o simbuild/micaz/pytosin.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DIDENT_T
IMESTAMP=0x6ac61fb3L -DIDENT_UIDHASH=0x05c16db5L -I../tos/lib/tossin/tossin_wrap.cxx -I/usr/include/python2.7 -I../tos/lib/tossin -DHAVE_CONFIG_H
../tos/lib/tossin/tossin_wrap.cxx: In function 'void SWIG_Python_AddErrorMsg(const char*)':
../tos/lib/tossin/tossin_wrap.cxx:880:42: warning: format not a string literal and no format arguments [-Wformat-security]
    PyErr_Format(PyExc_RuntimeError, msg);
                                         ^
g++ -c -shared -fPIC -o simbuild/micaz/pytosin.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DIDENT_T
IMESTAMP=0x6ac61fb3L -DIDENT_UIDHASH=0x05c16db5L -I../tos/lib/tossin/tossin.c -I/usr/include/python2.7 -I../tos/lib/tossin
g++ -c -shared -fPIC -o simbuild/micaz/c-support.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DIDENT_
TIMESTAMP=0x6ac61fb3L -DIDENT_UIDHASH=0x05c16db5L -I../tos/lib/tossin/hashtable.c -I/usr/include/python2.7 -I../tos/lib/tossin
[INFO] linking into shared object ./TOSSIMmodule.so
g++ -shared -fPIC simbuild/micaz/pytosin.o simbuild/micaz/sim.o simbuild/micaz/tossin.o simbuild/micaz/c-support.o -lstdc++ -o _TOSSIMmodule.so
[INFO] copying Python script interface TOSSIM.py from lib/tossin to local directory

[INFO] *** Successfully built micaz TOSSIM library.
aquib@ubuntu:~/tinys-main/apps/Blink$ python sim.py
0:0:0.488281260 DEBUG (1): LEDS: Led0 off.
0:0:0.976562510 DEBUG (1): LEDS: Led0 on.
aquib@ubuntu:~/tinys-main/apps/Blink$
```